

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000786		
	Version:	10.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	24-Apr-2018
Title:	SM DG57. Preventive maintenance			

DG-57 Service Manual

PREVENTIVE MAINTENANCE

Procedure description
This procedure provides guidelines on the Preventive Maintenance to be performed by Field Engineers to the instrument. It must be performed on a yearly basis and the set of parts to be replaced depends on the number of preventive maintenances that have been passed by an instrument. All parts described here can be filled in the Preventive Maintenance checklist (see PM_checklist.pdf in Annex folder). <i>Note: Document REGD-0000786_v10 replaces document REGD-0000786_v9.</i>

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EQUIPMENT

1. ---

SET UP

1. ---

PROCEDURE

Process	Expected Results
Initial checks 1. Visual inspection of the following parts: - General appearance: no dirt or scratches. - Serial Cable: no scratches or broken parts. - Serial Connector: correctly assembled, not damaged. - Video cable: no scratches nor broken parts. - Video connector: correctly assembled, not damaged.	Visual inspection should be correct
Software 2. Inspection of the following: - Software version: latest Diana System version should be installed. - Folders: all system folders should be available. - Register configuration: check all user parameters are set correctly. - Software setup: check that all user parameters are set correctly.	Software must be properly set
Parts replacement 3. Replace the following parts (depending on time and condition): - 231932 'Probe' or 232214 'Probe (for impact detector)' (every year) (see document REGD-0000828). - 231821 'Syringe' (every year) (see document REGD-0000831). - 232213 'Dilution station cup' (every year). - 232279 'Washing station v4 (with tube connection)' (every 2 years). - Tubing (Every 3 years) (see document REGD-0000822). i. Syringe input silicon tube: 231952 Silicone tube 3x1.5 (25 mts) ii. Syringe to Clot detector Teflon tube: 230126 Teflon tube 1.1x1.7 mm (5mts) iii. Clot detector to probe teflon tube. 230126 Teflon tube 1.1x1.7 mm (5mts) - Wiring (Every 3 years) (see document REGD-0000823). i. 232061 Gripper chain cables 108, 109. ii. 232060 Level detector chain cable 79. iii. 232042 Gripper&Probe motor encoder cables.	---

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Process	Expected Results
- Wiring and tubing (Every 3 years). i. 232190 X axis chain with wiring and PVC tubes. ii. 232191 Y axis chain with wiring and PVC tubes. - 232197 Pumps silencer (Every year). - Pressure and vacuum pump (Every 3 years) (see document REGD-0000829): i. 232237 Pressure pump 24V with PCB pressure regulator. ii. 232238 Vacuum pump 24V with PCB. Common Modules 4. Mechanics: check for smooth door movement. 5. Electronics: verify the following parts: - Power supply voltage 24V (see document REGD-0000791). - Conversion to 8V (see document REGD-0000792). - Conversion to 12V (see document REGD-0000792). - Status led, centrifuge access led. - Instrument initialization (if there is any problem, see document REGD-0000837). - Switch off and unplug the electrical appliance. Check with a Multimeter the electric continuity between the earth terminal of the IEC inlet appliance and these points (every year): i. Screw of the incubator block. ii. Screw of the centrifuge cover. - Switch off and unplug the electrical appliance. Check the integrity of the "Main Ground lug". See document REGD-000848. (Every year). Samples and Reagents Module 6. Mechanics: - Lubricate Samples carousel crown inner side (see document REGD-0000841). - Check Samples carousel crown tension. - Clean transparent film of home sensors. - Clean carousels. - Check spinning system. 7. Electronics: - Home cycles (10 minutes) (see document REGD-0000797). - Random movements (10 minutes) (see document REGD-0000797). - Reagents reading (see document REGD-0000815). - Samples reading –BC and diameter- (10, 12 and 15mm tubes) (see document REGD-0000815). Incubator Module 8. Mechanics: - Clean incubator. Centrifuge Module 9. Mechanics - General cleaning. - Card holders status and integrity. - Check the integrity of the rotor, looking for fissures or damages. (Every year).	Door movement should be smooth All test must pass successfully Continuity Continuity --- All test must pass successfully --- ---

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Process		Expected Results
10. Electronics:	- Home cycles (10 minutes) (see document REGD-0000804). - Random positioning (10 minutes) (see document REGD-0000804). - Unbalance sensor (only for SN<057-0004285, see document REGD-0000803).	All test must pass successfully
Robot Module		
11. Mechanics:	- Clean Robot head. - Lubricate Z leadscrew & wagon guide (see document REGD-0000841). - Lubricate leadscrew ball bearings (see document REGD-0000841). - Lubricate X and Y axis (see document REGD-0000841). - Check pulleys status. - Check belts status. - Check Gripper pins status. - Check Gripper fingers movement.	---
12. Electronics:	- Initialisation (see document REGD-0000805). - Homes (see document REGD-0000805). - Random movements (10 minutes) (see document REGD-0000805). - Cards transport (30 minutes) (see document REGD-0000805). - Level detection test (100 detections at different height) (see document REGD-0000805).	All test must pass successfully
Fluidic System		
13.	- Syringe test (see document REGD-0000821). - Bottles level detection (see document REGD-0000821). - Pumps (Performance and noise) (see document REGD-0000821). - Pressure/Vacuum (see document REGD-0000821).	All test must pass successfully
Reader Module		
14. Mechanics:	- Clean cards light diffuser: use an anti-static cleaner - Check led's performance: all led should light on. - Check camera cables and connector: they should be all in good condition.	---
15. Electronics:	- Fine adjustment of the reader camera optics (see points 21 to 24 in document REGD-0000812). - Reference image creation (see document REGD-0000814) - Cards reading (see document REGD-0000815).	All test must pass successfully
Functional checkings		
16.	- Perform the Operation Qualification (see document REGD-0007337)	---

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DOCUMENT CHANGE HISTORY

Version	Date	Change	Rationale
2.0	2005/05/09	Document REGD-0000786_v2 replaces document 57013V0M000.	Document reference changed.
3.0	2011/05/04	Added 'previous version 'data', 'change' and 'rationale' inside document change history section.	Update change history for previous document history modification.
3.0	2011/05/04	Updated the Document Change History: Date format changed to YYYY/MM/DD Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
3.0	2011/05/04	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
3.0	2011/05/04	Document REGD-0000786_v3 replaces document REGD-0000786_v2.	Document version changed.
3.0	2011/05/04	Updated 'parts replacement' section.	Added/modified spare part codes.
4.0	2011/05/06	Added on point 6: - Checking of the electrical continuity. - Integrity check of the "Main Ground lug".	Added according the standard "Safety requirements for electrical equipment for measurement, control, and laboratory use Part 1: General requirements (IEC 61010-1:2001), point 61010-2-020 (Equipment maintenance).
4.0	2011/05/06	Added on point 11: - Integrity check of the card holders. - Integrity check of the rotor.	Added according the standard "Safety requirements for electrical equipment for measurement, control, and laboratory use Part 1: General requirements (IEC 61010-1:2001), point 61010-2-020 (Equipment maintenance).
5.0	2011/05/23	Document REGD-0000786_v5 replaces document REGD-0000786_v4.	Document version changed.
5.0	2011/05/23	Header title.	Increase information to easy document search.
5.0	2011/05/23	Document title.	Commercial reference substituted by project reference.
6.0	2011/06/16	Document REGD-0000786_v6 replaces document REGD-0000786_v5.	Document version changed.
6.0	2011/06/16	Modified continuity check inside point 6.	Modified according the standard "Safety requirements for electrical equipment for measurement, control, and laboratory use Part 1: General requirements (IEC 61010-1:2001), point 61010-2-020 (Equipment maintenance).
7.0	2011/09/27	'+' sign replaced by '±' inside section 10.	Error corrected.
8.0	October 2011	'Parts replacement' section updated.	-Replaced 'other tubes' by the concrete tubes that has to be checked. -Replaced 'maximun' period of time by a concrete period of time.
9.0	December 2014	Some actions have been modified / deleted from the previous version	- Actions already required in the OQ (REGD-0007337) have been removed - Reader camera fine adjustment (and the corresponding reference image creation) is added
10.0	April 2018	Washing station replacement frequency	Washing station should be changed during PM every 2 years.

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